

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Double Blind, Placebo Controlled, Phase III Study to Evaluate the Efficacy and Safety of Anifrolumab in Adults With Chronic and/or Subacute Cutaneous Lupus Erythematosus Who Are Refractory and/or Intolerant to Antimalarial Therapy **Short Title/Acronym:** LAVENDER **Protocol Number:** D346BC00001 **NCT Number:** NCT06015737 **Sponsor Name:** AstraZeneca **Investigational Product:** Anifrolumab (SAPHNELO) / Subcutaneous (SC) injection **Phase of Development:** Phase III **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy and safety of subcutaneous (SC) anifrolumab versus placebo in adult participants with chronic and/or subacute cutaneous lupus erythematosus (CLE) who are refractory and/or intolerant to antimalarial therapy. **Secondary Objectives:** To evaluate the maintenance of a CLASI-70 response from Week 24 up to Week 52, to assess the pharmacokinetics (Ctrough) of subcutaneously administered anifrolumab, and to measure immunogenicity.

2.2 Background & Rationale To address the high unmet medical need and gap in effective therapeutic options for adult patients living with chronic and/or subacute Cutaneous Lupus Erythematosus (CLE) whose disease is refractory and/or intolerant to standard-of-care antimalarial therapies. This study investigates a subcutaneous (SC) administrative route, to potentially provide a more convenient patient experience compared to intravenous (IV) formulations.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 460 adult participants **Number of Sites:** Multicenter (Global) **Estimated Study Start:** June 2024 **Estimated Primary Completion:** August 2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age $\geq$ 18 to 70 years. 2. Confirmed diagnosis of CLE (clinically and histologically) with a Cutaneous Lupus Erythematosus Disease Area and Severity Index-Activity (CLASI-A) total score $\geq$ 10 points at Screening. 3. Inadequate response or intolerance to antimalarial therapy.

4.2 Exclusion Criteria 1. Severe or life-threatening Systemic Lupus Erythematosus (SLE) or active SLE / Sjögren's Syndrome. 2. History or evidence of suicidal ideation. 3. History of catastrophic antiphospholipid syndrome (APS) or recurrent infections requiring hospitalization and IV antibiotics.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Anifrolumab or matching Placebo administered via subcutaneous (SC) injection. **Route of Administration:** Subcutaneous (SC) **Duration of Treatment:** Up to 51 weeks total. Consists of a 24-week double-blind, placebo-controlled period (Week 0 to 23), followed by a 28-week open-label, uncontrolled treatment period (Week 24 to 51) where all participants receive anifrolumab.

5.2 Concomitant & Prohibited Therapy Permitted: Stable baseline treatments/standard of care for CLE, including specific topical calcineurin inhibitors or systemic glucocorticoids, based on protocol rules. **Prohibited:** Other biologic therapies, investigational agents, and unauthorized immunomodulatory treatments during the study timeframe.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, CLE diagnosis/Biopsy, TB screening, CLASI-A baseline
Baseline	Day 0	Randomization, First SC Dose (Anifrolumab/Placebo)
Interim 1	Week 24	Primary efficacy assessment (CLASI-70), transition to Open-Label Anifrolumab treatment
End of Study	Week 52 / 64	Final efficacy/safety assessments, completion of open-label period, 12-week safety follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Number of participants with a $\geq 70\%$ reduction relative to baseline in the Cutaneous Lupus Erythematosus Disease Area and Severity Index-Activity (CLASI-A) score (defined as a CLASI-70 responder) at Week 24. **Measurement**

Method: Clinical investigator assessment using the validated CLASI scoring instrument.

7.2 Secondary & Exploratory Endpoints * Number of participants maintaining a CLASI-70 response from Week 24 up to Week 52. * Pharmacokinetics (Ctrough) evaluation of SC anifrolumab. * Evaluation of immunogenicity in participants with chronic and/or subacute CLE.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via onsite visits, patient-reported symptoms (including monitoring for serious infections and injection-site reactions). **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness to the sponsor. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee assigned to review safety signals throughout the trial's double-blind and open-label periods.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy analysis; Safety Set for AE/SAE monitoring. **Sample Size Justification:** Approximately \$N=460\$ participants, powered to detect a statistically significant difference in CLASI-70 response rates at Week 24 between the anifrolumab and placebo groups. **Handling of Missing Data:** Non-Responder Imputation (NRI) typically applied for binary endpoints (CLASI-70 response) or Multiple Imputation for continuous metrics.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring visits for source data verification and assessment of protocol adherence. **Audit Trail:** Validated Electronic Data Capture (EDC) system for eCRF locking, query management, and data clarification.

11. Study Identifiers & Governance

Primary ID: D346BC00001 **Registry Number:** NCT06015737 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** LAVENDER **Official Regulatory Title:** A Multicenter, Randomized, Double Blind, Placebo Controlled, Phase III Study to Evaluate the Efficacy and Safety of Anifrolumab in Adults With Chronic and/or Subacute Cutaneous Lupus Erythematosus Who Are Refractory and/or Intolerant to Antimalarial Therapy

12. Clinical Framework (Cutaneous Lupus Erythematosus Specific)

Primary Condition: Chronic and/or Subacute Cutaneous Lupus Erythematosus (CLE) **Diagnosis Threshold:** CLASI-A total score ≥ 10 points **Medical Methodology:** Non-responders/intolerant to Antimalarial Therapy **Optimization Target:** Significant reduction in skin disease activity (CLASI-70 response)

13. Study Procedures & Methodology

Management Pathway: Subcutaneous (SC) biological therapy pathway **Education Protocol (HCP):** Standardization of CLASI instrument scoring and biopsy confirmation **Education Protocol (Patient):** SC injection self-administration and infection risk awareness **Quality Audit Mechanism:** Independent central review of dermatological metrics and routine data quality audits

14. Observation & Follow-Up Schedule

Total Duration: Up to 64 weeks (52-week treatment span + 12-week safety follow-up) **Onsite Visit Cadence:** Frequent early phase visits, spanning out to Weeks 12, 24, 52 **Remote Monitoring Frequency:** Routine data reviews per trial monitoring plan **Checklist Review Interval:** Major milestones assessed at Week 24 (End of double-blind) and Week 52 (End of open-label)

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				